

Chapter 15 One-Pager — Research Frontier Mapping and Contribution Design

Research on the possibilities for applying Artificial Intelligence in computer games

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Problem. Fourteen chapters built a toolbox; a thesis needs three defensible claims. The April 2026 plan overclaimed. Real-time opponent modelling already works in two-player games (GSCU, StratFormer); one benchmark already combines return with exploitability; equal share is not a guarantee; piKL anchors to a human policy, not to Nash. Only C2 is open at its core: exploitation with a checked loss bound beyond two players, tested against colluders.

Approach. A frontier map per contribution (what exists, the verified gap, the approach, evidence, risks); design documents for C1–C3; six experiment specifications; a six-paper pipeline. Two pilots were run on Chapter 14’s exact engine. **P1:** Ganzfried and Sandholm’s RWYWE, which risks only the gains it can prove the opponent gave, replicated against their table and measured with Chapter 14’s protocol. **P2:** three-player Kuhn, with three bounded deviation rules against independent, fixed-colluding and adaptively colluding pairs. All numbers are measured, 5–10 seeds.

Key results (measured).

- *The contributions, as they may now be claimed.* C1: within-match inference for N players, coupled to a safety criterion. C2: exploitation beyond two players with a checked loss criterion; no general theorem, small games, equal share as a reference line. C3: a joint protocol that catches what existing measures miss, plus match-level safety.
- *RWYWE replicates where it matters.* All safe algorithms stay above the game value in 400 of 400 matches against an opponent that best-responds every hand. The best-response row is not reproduced (0.328 vs 0.470; open).
- *RWYWE is the two-player baseline Chapter 8 lacked.* Its gain over the blueprint is +0.062 (Kuhn) and +0.113 (Leduc), $3.1\times$ and $2.4\times$ the best equilibrium’s. It stays safe over the match in all 120 attacked matches, but gains only 14–30 % of what RNR(0.5) gains with no guarantee. Proving gifts is the bottleneck: with cards seen only at showdown, its Leduc bank stays at 0.01–0.02 chips.
- *Safety is a match-level property.* RWYWE spends a 2.19-chip bank within ~ 100 hands of an attack while the match stays above v^* ($S = +0.112$). C3 adds this readout.
- *Three players, exact checks.* The team-maxmin value of three-player Kuhn is -0.0076 per seat on average, close to equal share. Ganzfried and Sandholm’s bank with that floor gains +0.082 against independent pairs. It keeps -0.007 under adaptive colluders, where the blueprint gets -0.119 , and is safe in all 135 matches. A capped mixture holds its per-hand relative bound exactly, but gains only 16 % of the unbounded exploiter’s gain at $\epsilon = 0.1$.
- *Two robustness readouts disagree.* A KL anchor loses to the mixture on worst-case loss relative to the blueprint, but keeps far more against an adaptive coalition (-0.286 vs -0.379).
- *A defect found.* LP ties returned a degenerate equilibrium that never bluffs and never bets the King; breaking ties toward the blueprint fixed it.

Thesis connection. This is the bridge from Chapter I to Chapters II–IV. The goal and thesis of § 1.7 stand. RQ1 now asks for calibrated inference, RQ2 for relative and absolute bounds, RQ3 for match-level safety. The first publication is the joint protocol paper for IEEE CoG 2027 (1 March 2027), with AAMAS 2027 as a stretch.

Open questions. Within-hand gift detection to make gifts provable; learned coalition exploiters for three-player Leduc, calibrated on three-player Kuhn; whether the maximin value stays close to equal share in larger games; the unexplained best-response gap in the replication.